

Emily E. Hayden

Postdoctoral Fellow
Climate and Global Dynamics (CGD) Laboratory, NSF NCAR
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EDUCATION

2025	PhD Ocean and Atmospheric Physics (3.92 GPA)	<i>CEOAS, Oregon State University</i>
2021	MSc Ocean and Atmospheric Physics (3.89 GPA)	<i>CEOAS, Oregon State University</i>
2018	BA Physics, <i>summa cum laude</i>, Dean's List (3.83 GPA)	<i>University of Colorado - Boulder</i>
2015	AS Physics & AA General Studies, President's List (4.0 GPA)	<i>Colorado Mountain College</i>

RESEARCH APPOINTMENTS

2026-Present	Postdoctoral Research Fellow - NSF NCAR Fellow of the Advanced Study Program (ASP) Hosted by the Climate and Global Dynamics (CGD) Lab
2021-2025	Doctoral Research Fellow - CEOAS Dissertation: <i>Characterizing the Role of the Atmosphere in Bering Sea Temperature Variability</i> Advisors: Dr. Larry W. O'Neill and Dr. Seth F. Zippel
2019-2021	Graduate Research Fellow - CEOAS Project: <i>Evaluating the Role of Air-Sea Interactions in Bering Sea Warming</i> Advisor: Melanie R. Fewings
2016-2018	Undergraduate Honors Researcher - Soft Materials Research Center Honors Thesis: <i>Liquid Crystal Phases of RNA Mononucleoside</i> Advisor: Dr. Noel A. Clark
2017-2018	Undergraduate Researcher - JILA Physics Frontiers Center Project: <i>Personalization of Learning in an Inquiry-based Environment</i> Advisors: Dr. Noah Finkelstein and Dr. Mike Bennett
2017	Undergraduate Researcher - Soft Materials Research Center Project: <i>Liquid Crystal Phases of DNA and RNA</i> NSF Research Experience for Undergraduates (REU) Program

TEACHING AND MENTORING

2026	Isha Khandwala – Rising UW Graduate Research Fellow Science and research mentor as part of NSF SOARS
2023-2024	Helena Rocha – Undergraduate Honors Researcher Co-advised with Dr. Laurie Juranek and Dr. Miguel Goñi
2022-2024	Tyanna Fox – Undergraduate Researcher Mentored with Dr. Larry W. O'Neill
2022	Tyanna Fox – Undergraduate Climate Science Student Academic and development mentor as part of GUIDE
2021-2023	Ella Carpenter – Undergraduate Researcher Mentored as part of the NSF ARC-Learn Program.
2021-2023	Kacper Kardasz – Undergraduate Researcher Mentored as part of the NSF ARC-Learn Program.
2021-2023	Helena Rocha – Undergraduate Researcher Mentored as part of the NSF ARC-Learn Program.

- 2021 **Claire Williams** – Undergraduate Climate Science Student
Academic and development mentor as part of GUIDE
- 2021 **Teaching Assistant**
ATS 201: *Introduction to Climate Science*
- 2020 **Saray Valdez** – Undergraduate Climate Science Student
Academic and development mentor as part of GUIDE

PUBLICATIONS

Peer-Reviewed Articles

- Hayden, Emily E.**, Seth F. Zippel, Justin J. Wettstein, and Larry W. O'Neill. Cross-Timescale Air-Sea Coupling in the Bering Sea. *Journal of Climate*, 39 (13), 3931-3949. 10.1175/JCLI-D-25-0562.1
- Hayden, Emily E.**, Larry W. O'Neill, and Seth F. Zippel (2024). Atmospheric Variability Drives Anomalies in the Bering Sea Air–Sea Heat Exchange, *Journal of Climate*, 37(24), 6659-6678. 10.1175/JCLI-D-24-0105.1.
- Hayden, Emily E.** and Larry W. O'Neill (2024). Processes Contributing to Bering Sea Temperature Variability in the Late 20th and Early 21st century, *Journal of Climate*, 37(1), 41-58. 10.1175/JCLI-D-23-0331.1.
- Smith, Gregory P., Tommaso P. Fraccia, Marco Todisco, Giuliano Zanchetta, Chenhui Zhu, **Emily E. Hayden**, Tommaso Bellini, and Noel A. Clark (2018). Backbone-free duplex-stacked monomer nucleic acids exhibiting Watson-Crick selectivity. *Proceedings of the National Academy of Sciences*, 115(33), E7658-E7664. 10.1073/pnas.1721369115. *Recipient of the 2018 Cozzarelli Prize.*

White Papers

- Nuijens, Louise, Jacob Wenegrat, and others including **Emily E. Hayden** (2024). The air-sea interaction (ASI) submesoscale: Physics and impact. 10.5065/78ac-qd31.

FELLOWSHIPS AND AWARDS

- 2025 **ASP Postdoctoral Fellowship** – NSF NCAR
- 2025 **CICOES Postdoctoral Fellowship** – University of Washington and NOAA PMEL (*declined*)
- 2025 **NOAA C & G.C. Postdoctoral Fellowship** – NSF NCAR (*program cancelled*)
- 2023 **Exemplary Presentation Award (Honorable Mention)** – NDSEG Fellows Conference
- 2023 **Wayne V. Burt Award** – CEOAS Award recognizing outstanding scientific achievement
- 2023 **Scholarly Presentation Award** – OSU Graduate School
- 2021 **NDSEG Fellowship** – United States Department of Defense, Office of Naval Research
- 2021 **FINESST Fellowship** – NASA (*declined in favor of NDSEG*)
- 2019 **Provost Distinguished Graduate Fellowship** – OSU Graduate Fellowship
- 2019 **Prestigious Diversity Fellowship** – OSU Graduate Fellowship (*declined in favor of Provost*)
- 2017 **Moehrke Educational Advancement Scholarship** – University of Colorado, Boulder
- 2016 **Latino Alumni Association Scholarship** – University of Colorado, Boulder
- 2015 **University Women's Club Scholarship** – University of Colorado, Boulder
- 2015 **Rothberg Community College Scholarship** – University of Colorado, Boulder
- 2015 **Louise L. Noble Scholarship** – University of Colorado, Boulder
- 2015 **First Generation Grant** – University of Colorado, Boulder
- 2015 **White Rose Foundation Scholarship** – University of Colorado, Boulder
- 2015 **Wayne Swanson Scholarship** – University of Colorado, Boulder

INVITED TALKS

- Jan, 2025 **Characterizing the role of atmospheric circulation in Bering Sea variability and extremes**
US Naval Research Laboratory (NRL) Seminar, *Stennis Space Center, MS*
- Nov, 2022 **Atmospheric Circulation Anomalies and their role in recent Bering Sea Warming**
NOAA - PMEL EcoFOCI 2022 Fall Seminar Series, *Seattle, WA*

PRESENTATIONS

- Sep, 2026 **Mesoscale Ocean Impacts on the Midlatitude Storm Tracks on S2D Timescales**
CLIVAR Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop – Boulder, CO
- Sep, 2026 **Improving the Predictability of Midlatitude Storms Through a High-Resolution Study of Air-Sea Coupling**
WCRP ESMO Third Annual S2S2D Conference – Reading, UK (Poster)
- Feb, 2026 **Identifying Cross-Timescale Coupling in Bering Sea ASI Using Bispectral Statistics**
Ocean Sciences Meeting 2026 – Glasgow, UK (Poster)
- Feb, 2025 **Characterizing the Role of Atmospheric Circulation in Bering Sea Thermal Variability**
CLIVAR Climate Dynamics Workshop on Weather-Climate Interactions – Melbourne, AUS (Poster)
- Jan, 2025 **Characterizing the Role of Atmospheric Circulation in Bering Sea Thermal Variability**
AMS Annual Meeting 2025 – New Orleans, LA
- Oct, 2024 **Characterizing the role of the atmosphere in Bering Sea temperature variability**
OSU Physics of Oceans and Atmospheres Seminar – Corvallis, OR
- Feb, 2024 **Identifying the atmospheric processes that contribute to Bering Sea ocean temperature variability**
Ocean Sciences Meeting 2024 – New Orleans, LA (Poster)
- Aug, 2023 **Bering Sea warming in the late 20th and early 21st century** (Exemplary Presentation – Honorable Mention)
NDSEG Fellows Conference – San Antonio, TX
- Mar, 2023 **Atmospheric processes contributing to Bering Sea warming in the late 20th and early 21st century**
US CLIVAR Mesoscale and Frontal-Scale Air-Sea Interactions Workshop – Boulder, CO (Poster)
- Dec, 2022 **Bering Sea Extremes: Atmospheric Circulation Variability drives Anomalies in Air-Sea Heat Fluxes**
AGU Fall Meeting 2022 – Chicago, IL (Poster)
- Jun, 2022 **Variability in Atmospheric Circulation over the Bering Sea and its Impact on Air-Sea Heat Fluxes**
US CLIVAR Circulation of the Arctic and Sub-Arctic Seas Workshop – Seattle, WA
- May, 2022 **Bering Sea Atmospheric Circulation Variability and Air-Sea Heat Fluxes**
International Ocean Vector Winds Science Team Meeting – Virtual (Poster)
- Mar, 2022 **Variability in Atmospheric Circulation over the Bering Sea and its Impact on Air-Sea Heat Fluxes**
Ocean Sciences Meeting 2022 – Virtual
- Feb, 2021 **Evaluating the role of air-sea interactions in Bering Sea warming**
International Ocean Vector Winds Science Team Meeting – Virtual (Poster)
- Aug, 2018 **Personalization of Learning in an Inquiry-based Environment**
Physics Education Research Conference – Washington, DC (Poster)
- Aug, 2017 **Liquid Crystal Phases of RNA Mononucleosides**
REU Final Review – Boulder, CO (Poster)
- Jul, 2017 **Liquid Crystal Phases of DNA and RNA**
International Society for the Study of the Origin of Life Conference – San Diego, CA (Poster)
- Apr, 2017 **Liquid Crystal Phases of RNA Mononucleoside Triphosphates**
University of Colorado Undergraduate Physics Poster session – Boulder (Poster)

PROFESSIONAL ACTIVITIES

- 2026-Present **Research Review Committee Chair**
NSF NCAR Advanced Study Program (ASP)
- 2025-Present **Ad-hoc referee for scientific journals**
Reviewer for *Journal of Climate*, *Climate Dynamics*, *Frontiers in Marine Science*, and *GRL*
- 2024-2025 **OASIS Task Team Member**
Observing Air-Sea Interactions Strategy (OASIS) Satellite Air-Sea Fluxes: New Views from Space
- Jan, 2025 **AMS Annual Meeting 2025 Session Co-Chair**
Served as a co-chair for the *Air-Sea Interactions at Climate Scales* session – New Orleans, LA
- Feb, 2024 **Ocean Sciences 2024 Session Co-Chair**
Served as a co-chair for the *Oceanic and Atmospheric Interactions and Drivers: Their Scale Dependence and Climate Impacts* session – New Orleans, LA

SUMMER SCHOOLS AND WORKSHOPS

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- 2026 **Broadening Earth System Research Across Scales** **US CLIVAR – Boulder, CO**
A CLIVAR workshop focused on recent advances in high-resolution (HR) climate models, with an emphasis on identifying the scientific value of these model, coordinating future HR modelling efforts, and determining best practices to ensure open access to these simulations.
- 2026 **CESM Workshop** **NCAR – Boulder, CO**
An NSF NCAR annual workshop focused the Community Earth System Model (CESM), with an emphasis on advancing model development and research applications, and fostering collaborations and knowledge exchange between CESM scientists and the broader science community.
- 2026 **High and Ultra-High Res Modeling of the Earth System** **Texas A & M – College Station, TX**
A workshop focused on an assessment of current generation high-resolution ocean, atmosphere, and coupled Earth system models, and on identifying next steps for improving their their predictive capacity.
- 2024 **CESM Tutorial 2024** **NCAR – Boulder, CO**
An NSF NCAR Community Earth System Model (CESM) Tutorial designed to provide attendees with a scientific understanding of model components, and skills for running the model.
- 2024 **Ocean Uncertainty Quantification Summer School** **CLIVAR – Miami, FL**
A CLIVAR summer school that aimed to teach best practices for improving the derivation, interpretation, and utilization of ocean data uncertainties to the next generation of scientists.
- 2024 **Observing Air-Sea Interactions Strategy Workshop 2024** **OASIS – New Orleans, LA**
A face-to-face meeting of the OASIS community, with the goal of reviewing the OASIS mission, current activities, and projects.
- 2023 **Atmosphere-ocean coupling at (sub)mesoscales** **Lorentz Center – Leiden, NL**
A workshop convening an international group of scientists in the field of atmospheric and oceanographic sciences. Resulted in co-authorship on a white paper.
- 2023 **Mesoscale and Frontal-Scale Air-Sea Interactions Workshop** **US CLIVAR – Boulder, CO**
A CLIVAR workshop focused on identifying gaps in understanding, facilitating, and improving comparison of observations and modelling, and to build collaborations within the air-sea interactions community.
- 2022 **Circulation of the Arctic and Sub-Arctic Seas Workshop** **US CLIVAR – Seattle, WA**
A CLIVAR workshop focused on an assessment of current understanding of Arctic Ocean circulation and discussion on observing system requirements. *Oral Presentation.*
- 2020 **Satellite Observations and Climate Models Summer School** **NASA – Pasadena, CA**
An annual summer school focused on "Using Satellite Observations to Advance Climate Models" (*Virtual due to COVID-19*).
- 2020 **AI for Earth System Science Summer School (AI4ESS)** **NCAR – Boulder, CO**
A weeklong course describing topics in artificial intelligence, machine learning, and deep learning and their relevance to Earth Science (*Virtual due to COVID-19*).

OUTREACH AND ENGAGEMENT

- 2026 **NSF SOARS:** Mentored a student pursuing air-sea interactions and high-resolution modelling research in the NSF NCAR CGD Laboratory. SOARS (Significant Opportunities in Atmospheric Research and Science) is a 10-week NSF-funded program, designed as an undergraduate-to-graduate atmospheric and Earth system science bridge program.
Role: Primary Science Mentor
NSF NCAR – Boulder, CO
- 2021-2023 **NSF ARC-Learn:** Mentored three undergraduate students in their Arctic research. ARC-Learn (Authentic Research for Collaborative Learning) was a 20-month NSF-funded program, designed as a mentorship program connecting undergraduate students from groups underrepresented in the Arctic sciences with faculty and graduate mentors in the field.
Role: Inclusive Mentoring Fellow
CEOAS – Oregon State University

- 2020-2022 **GUIDE: Graduate-Undergraduate Inclusive Development Experience:** Formerly known as AMP: Academic Mentoring Program. Mentored first-generation and underrepresented undergraduate students in their transition to graduate school. GUIDE was a 1-year program that paired rising OSU seniors with graduate-student mentors, with the goal of assisting mentees to learn about potential academic and career pathways.
Role: Undergraduate Mentor
CEOAS – Oregon State University
- 2022 **LSAMP: Louis Stokes Alliance for Minority Participation:** The College of Earth, Ocean, and Atmospheric Sciences at OSU organizes an annual, September tour for incoming LSAMP students to introduce them to academic and research opportunities in Earth Science. Assisted in coordinating and administering two science activity sessions for groups of incoming undergraduate students.
Role: Science Activity Lead
CEOAS – Oregon State University
- 2020-2021 **CEOAS Association of Graduate Students (CAGS) Discipline Representative:** Liaised between Physical Oceanography and Atmospheric Science graduate students and Deans and faculty, through the CAGS, with the primary goal of voicing student concerns and advocating for student interests.
Role: Physical Oceanography and Atmospheric Science Discipline Representative
CEOAS, Oregon State University
- 2015-2016 **Lego Robotics After-School Program:** Fostered elementary students to develop, build, and code a Lego robot to enter it in a regional competition.
Role: Science Activity Lead
Columbine Elementary School, Boulder, CO

PROFESSIONAL AFFILIATIONS

- 2025-Present **Mentoring Physical Oceanography Women to Increase Retention (MPOWIR)**
- 2020-Present **American Geophysical Union (AGU) Member**
- 2020-Present **American Meteorological Society (AMS) Member**
- 2019-2021 **Coalition of Graduate Employees (CGE) - Environmental Justice Caucus Member**
- 2016-2018 **Undergraduate Women in Physics (UWiP) Member**
- 2013-2015 **Phi Theta Kappa Member**